A Dynamic Model of Aggregate Supply and Demand

Brian C. Jenkins

University of California, Irvine

June 6, 2014

Equilbrium

- The full set of equilibrium conditions:

$$
\begin{aligned}
y_t &= E_t y_{t+1} - (r_{t+1} - \bar{r}) + g_t && (1)\\
\pi_t &= (1-\beta)\pi^* + \beta E_t \pi_{t+1} + \kappa(y_t - \bar{y}) + u_t && (2)\\
r_{t+1} &= i_{t+1} - E_t \pi_{t+1} && (3)\\
i_{t+1} &= \bar{r} + \pi^T + \phi_\pi(\pi_t - \pi^T) + \phi_y(y_t - \bar{y}) + v_t && (4)
\end{aligned}
$$

- Note that there are 4 equations and 4 endogenous variables: y_t, π_t, i_t, and r_t.

Table 1 : Calibrated parameter values for the dynamic AS-AD model.

Parameter	**Value**
$\bar{y}$	100
$\bar{r}$	2
β	0.98
κ	0.25
π^T	2
ϕ_π	1.5
ϕ_y	0

Demand Shock

- The solution for y_t and π_t when only shocks to the IS equation are present:

$$y_t = \bar{y} + \frac{1 - \beta\rho_g}{(1-\rho_g)(1-\beta\rho_g) + \kappa(\phi_\pi - \rho_g)} g_t \tag{5}$$

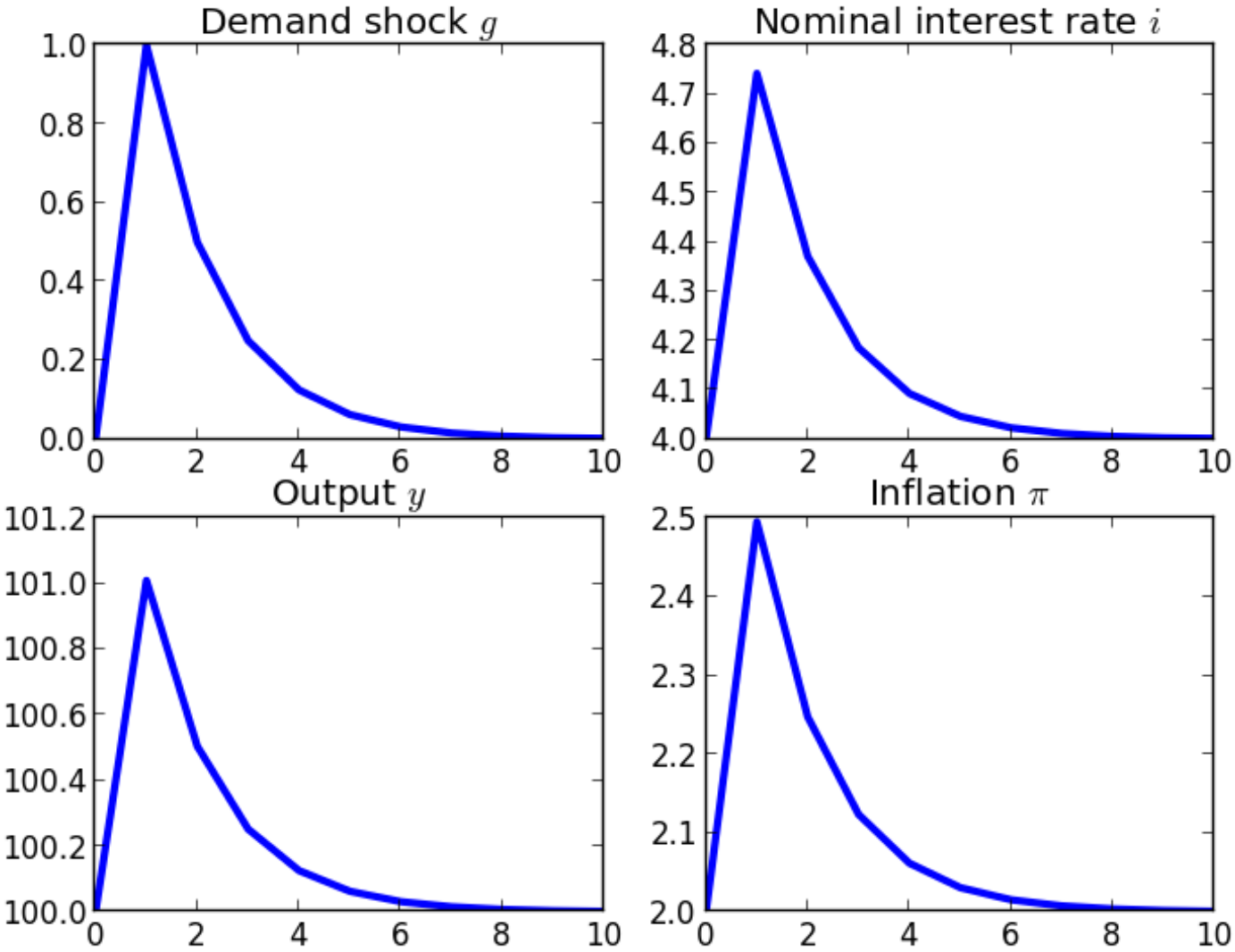
and:

$$\pi_t = \pi^T + \frac{\kappa}{(1-\rho_g)(1-\beta\rho_g) + \kappa(\phi_\pi - \rho_g)} g_t \tag{6}$$

- And the specification for g_t:

$$g_t = \rho_g g_{t-1} + \epsilon_t^g \tag{7}$$

Figure 1 : **Demand shock**. The responses of the nominal interest rate, output, and inflation to an exogenous 1% increase in the demand for goods and services. All vertical axis units are percent per year.



Inflation Shock

- The solution for y_t and π_t when only shocks to the supply equation or are present:

$$y_t = \bar{y} - \frac{\phi_\pi - \rho_u}{(1-\rho_u)(1-\beta\rho_u) + \kappa(\phi_\pi - \rho_u)} u_t \tag{8}$$

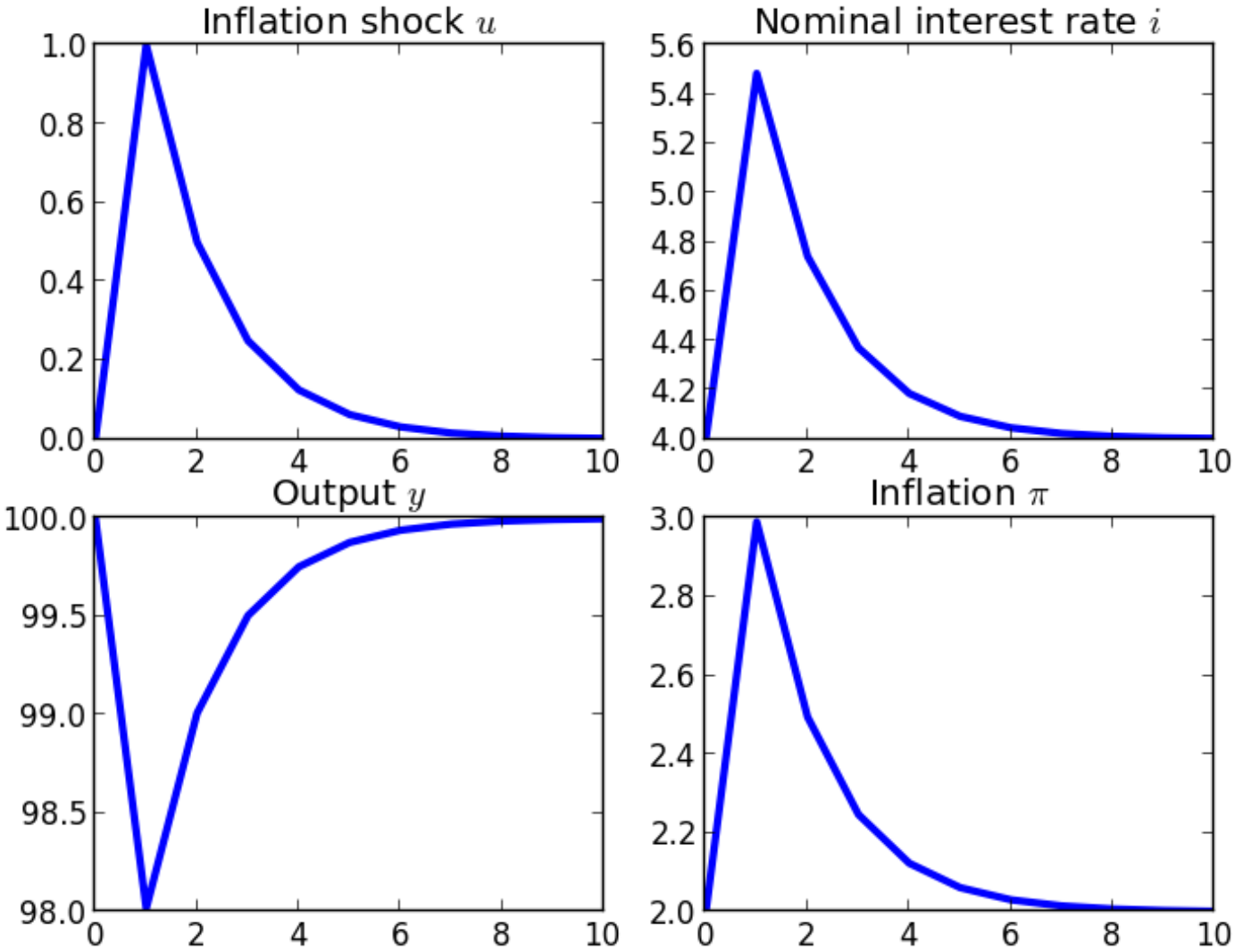
and:

$$\pi_t = \pi^T + \frac{1-\rho_u}{(1-\rho_u)(1-\beta\rho_u) + \kappa(\phi_\pi - \rho_u)} u_t \tag{9}$$

- And the specification for u_t:

$$u_t = \rho_u u_{t-1} + \epsilon_t^u \tag{10}$$

Figure 2 : **Inflation shock**. The responses of the nominal interest rate, output, and inflation to an exogenous 1% increase in the rate of inflation. All vertical axis units are percent per year.



Demand Shock

- The solution for y_t and π_t when only shocks to the monetary policy rule are present:

$$y_t = \bar{y} - \frac{1 - \beta\rho_v}{(1-\rho_v)(1-\beta\rho_v) + \kappa(\phi_\pi - \rho_v)} v_t \quad (11)$$

and:

$$\pi_t = \pi^T - \frac{\kappa}{(1-\rho_v)(1-\beta\rho_v) + \kappa(\phi_\pi - \rho_v)} v_t \quad (12)$$

- And the specification for g_t:

$$v_t = \rho_v v_{t-1} + \epsilon_t^v \quad (13)$$

Figure 3 : **Monetary policy shock**. The responses of the nominal interest rate, output, and inflation to an exogenous 1% increase in the nominal interest rate. All vertical axis units are percent per year.

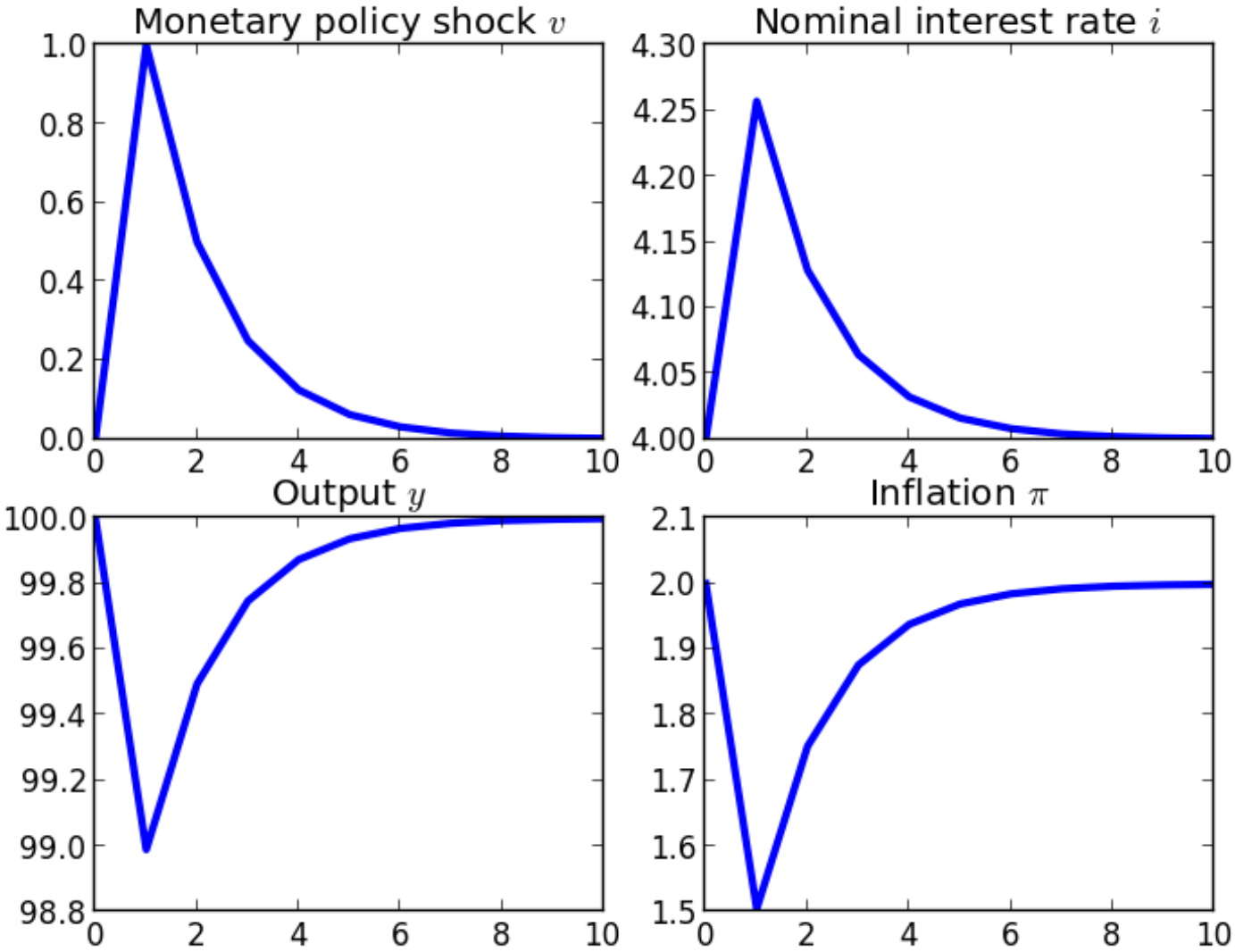


Figure 4 : **Demand shock**. The responses of the nominal interest rate, output, and inflation to an exogenous 1% increase in the demand for goods and services for increasingly greater weights on inflation in the monetary policy rule. All vertical axis units are percent per year.

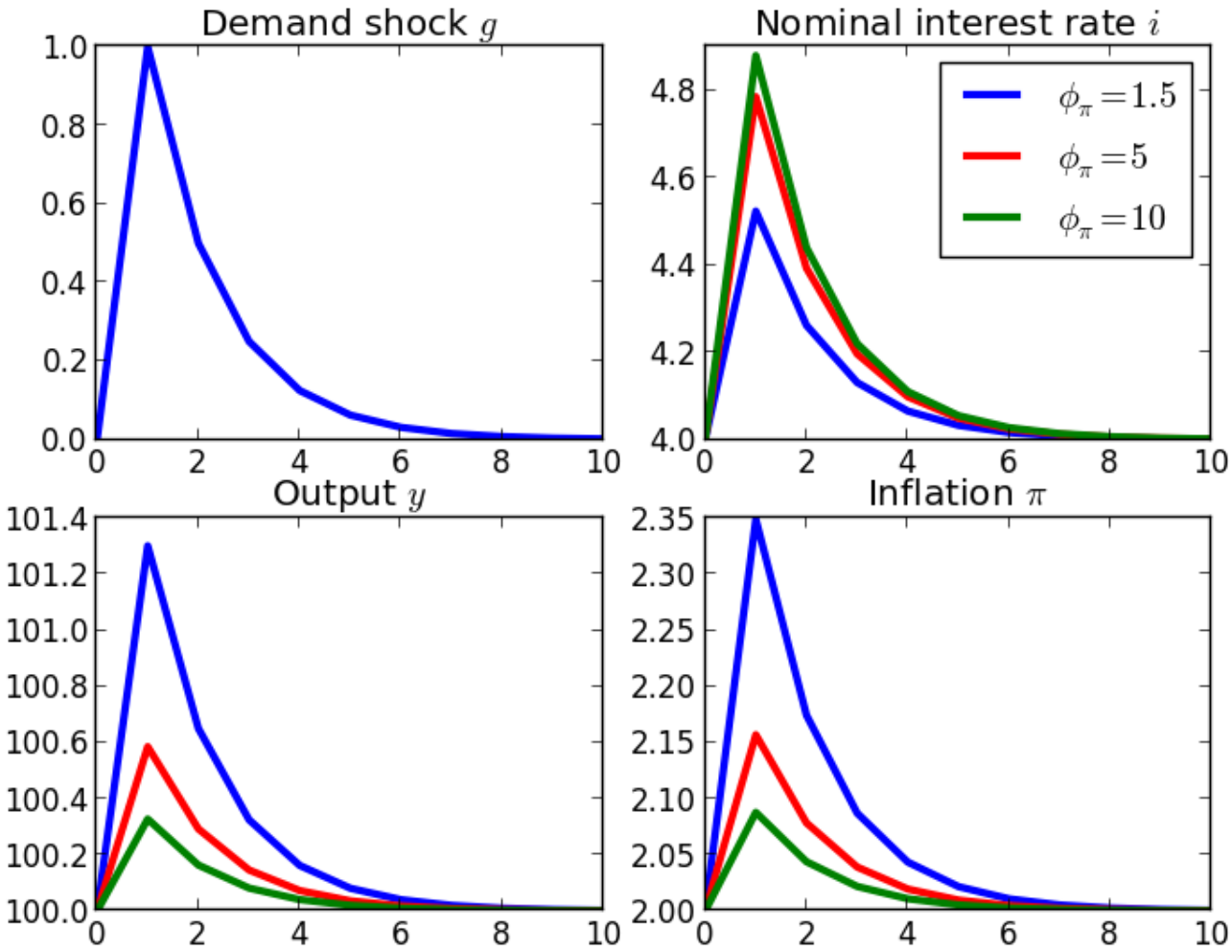
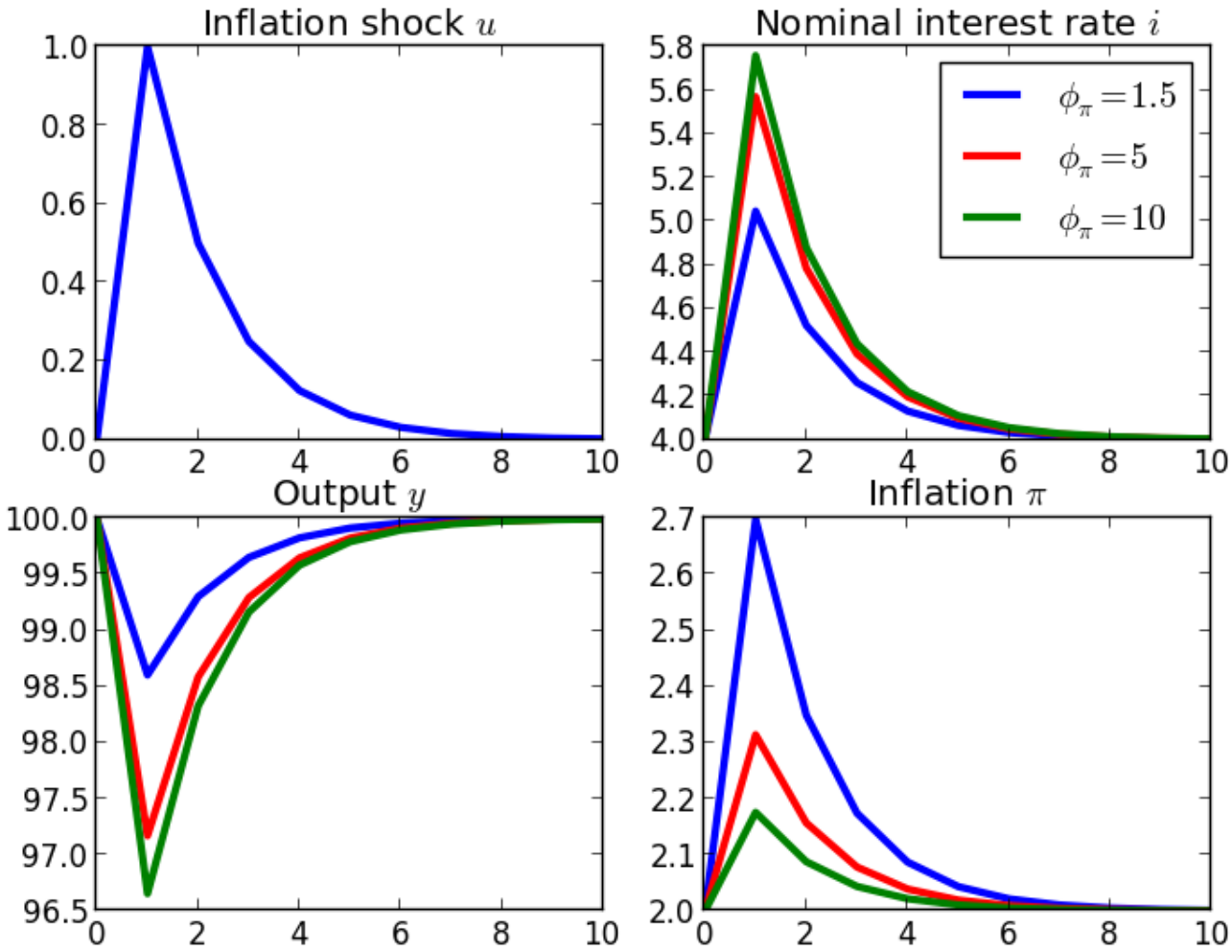



Figure 5 : **Inflation shock**. The responses of the nominal interest rate, output, and inflation to an exogenous 1% increase in the rate of inflation for increasingly greater weights on inflation in the monetary policy rule. All vertical axis units are percent per year.